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Occupational Exposure to AI: A Patent-Based Approach Using Large Language Models

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Abstract

Artificial Intelligence (AI) is reshaping labor markets, affecting occupations through automation and new skill requirements. Following Webb (2020), this study quantifies occupational exposure to AI using a patent-based approach combined with large language models (LLMs) and International Standard Classification of Occupations (ISCO-08) data. By leveraging semantic similarity analysis, we systematically map AI-related patents to specific job categories. Our findings indicate that ICT-related professions — Software Developers, Database Professionals, and ICT Service Managers — exhibit the highest AI exposure. However, AI's impact is expanding beyond technical roles, increasingly affecting managerial, creative, and process automation occupations. Temporal analysis reveals an expansion of AI adoption, initially concentrated in technical sectors (2000s), later extending to business automation (2010s), and more recently influencing creative industries (2020s). These results highlight the need for adaptive workforce policies, including reskilling programs and AI governance frameworks, to balance AI-driven productivity gains with labor market stability.

Keywords: *Artificial Intelligence; Patent Analysis; Large Language Models; Labor Market; Statistical Indicators; Time Series.*

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1. Introduction

The rapid advancements in Artificial Intelligence (AI) are reshaping industries, job roles, and skill requirements. AI-driven automation, machine learning, and large language models (LLMs) are increasingly embedded in professional tasks, raising concerns about job displacement, skill polarization, and wage inequality (Cazzaniga et al., 2024; OECD, 2021). The OECD (2021) notes that AI adoption disproportionately affects routine-based occupations while fostering growth in knowledge-intensive sectors. Similarly, Cazzaniga et al. (2024) highlight AI's role in amplifying labor market inequalities, benefiting high-income workers while middle-skill workers face substitution risks. Recent research (Comunale & Manera, 2024) suggests AI's impact extends beyond routine tasks to high-skill, cognitive-intensive jobs.

Efforts to quantify occupational exposure to AI have grown, with Webb (2020) introducing a patent-based approach to measure AI's labor market impact. This methodology, using patent data as a proxy for innovation, has been extended in studies like Dalla Zuanna et al. (2024), which find AI exposure concentrated in high-income, service-sector jobs. Cazzaniga et al. (2024) further show that advanced economies, with more cognitive-intensive occupations, are more exposed to AI and positioned to benefit. This paper employs a patent-based framework to quantify occupational AI exposure, mapping patents to job categories using the International Standard Classification of Occupations (ISCO-08). This study introduces an innovative approach that uses LLMs to improve semantic similarity analysis between patent abstracts and job descriptions, utilizing embedding-based techniques. The paper is structured as follows: Section 2 presents data sources, including Google Patents and ISCO-08 classifications. Section 3 details the methodology, covering patent retrieval, occupational classification, and similarity analysis. Section 4 discusses results, ranking AI-exposed occupations and analyzing trends. Section 5 concludes with policy recommendations to mitigate AI-driven labor disruptions.

2. Data

This section presents the data sources used in the study, detailing their origin, structure, and relevance to assessing occupational exposure to AI based on patent literature. The primary data sources are Google Patents and the ISCO-08. Google Patents provides a rich textual repository of global innovations. ISCO-08 serves as a comprehensive taxonomy of occupations, enabling alignment between patent content and job profiles.

2.1. Google Patents

Google Patents is a comprehensive digital platform offering open access to millions of patent documents from patent offices worldwide. It includes both granted patents and published applications from key jurisdictions such as the United States Patent and Trademark Office (USPTO), the European Patent Office (EPO), and the World Intellectual Property Organization

(WIPO), among others. It provides metadata on patent publications, including title, abstract, publication number, filing and publication dates, and patent office. Additional tables include assignees (names and locations), inventors, CPC codes for technology classification, citations for prior art references, and family members, which link related patents across jurisdictions. Patent document structures vary by country, and there is no universal format. However, some sections are common across most patents.

Table 1. Descriptions of the patent sections.

Section	Description
Title	The name of the invention
Abstract	A brief summary of the invention
Background	Context and prior art related to the invention
Summary	An overview of the invention and its advantages
Detailed Description	A comprehensive explanation of the invention
Claims	The legal scope and protection defined for the invention
Drawings/Figures	Visual representations of the invention

For this study, the abstract section was selected as the primary text input for the LLMs. Abstracts offer a standardized and condensed synthesis of each patent's purpose and technical scope, facilitating efficient comparison with occupational descriptions. Moreover, using abstracts helps mitigate noise from extremely detailed or inconsistent textual structures in other patent sections.

2.2. International Standard Classification of Occupations (ISCO-08)

The International Standard Classification of Occupations (ISCO-08) is a hierarchical framework developed by the International Labour Organization (ILO) to classify and group jobs based on their tasks and duties. It is a global reference for statistical and analytical purposes, facilitating international comparisons and harmonizing occupational data. Regarding the hierarchy, ISCO-08 is a system with four levels of aggregation: Major (10 categories), Sub-major (43 categories), Minor (130 categories), and Unit (436 categories).

The dataset used contains the following information:

Table 2. Descriptions ISCO classification.

Section	Description
Level	Level of the hierarchy (1 to 4)
Code	4-digit code which uniquely identifies the occupation class
Title	Title of the occupation
Definition	Definition, which describes the activity in more detail
Tasks include	Description of the activities included in the occupation class
Included occupations	Examples of jobs included
Excluded occupations	Examples of jobs not included to avoid class overlapping
Notes	Other related information to help the occupation definition

This study primarily uses the Minor group (Level 3) classification, as it strikes a balance between granularity and interpretability. While higher-level categories (e.g., Major groups) may be too broad to capture AI-related impacts, Unit groups (Level 4) can be overly specific, leading to sparse data issues.

3. Methodology

This section outlines the methodological framework adopted to extract occupational information from patent texts using LLMs. The process consists of three main phases: (3.1) patent data collection, (3.2) preparation of occupational descriptions from ISCO-08, and (3.3) semantic matching of patents to occupations via text embeddings. Each phase is described in detail below.

3.1. Patent Data Collection

Patent data were collected using Google Patents to analyze the extent to which different occupations are exposed to AI-related innovations. To this end, the following filters were applied during the retrieval process.

- *Keywords*: "artificial intelligence", "neural network", "llm", "large language model", "deep learning", "machine learning", "natural language processing"
- *Country*: European and American countries
- *Time range*: 2000-01-01 to 2024-10-30
- *Status*: grant, only granted patents
- *Language*: English, possibly translated with Google Translate provider

This query yielded **103,554 distinct patents**. A data scraping process was initiated to extract textual information on which to base the AI exposure analysis. Only the abstract section was retained.

3.2. Structuring Occupational Descriptions

The ISCO-08 was employed to align patent texts with job profiles. Its taxonomy offers a hierarchical structure of occupations consisting of four aggregation levels: Major, Sub-major, Minor, and Unit groups. As explained before, we operated at the **Minor group level (Level 3)**, which includes 130 occupational categories. Each class was enriched by **concatenating three textual components**: the title of the Minor group, the definition provided at Level 3 and definition of the corresponding Level 4 (Unit group) occupations, where available. This strategy provided a rich and semantically dense representation of each occupational category, suitable for downstream embedding tasks.

3.3. Embedding Models and Similarity Matching

We used embedding-based models to compute semantic similarity between patent abstracts and occupational descriptions, leveraging transformer architectures. The core model, **BGE-M3** (Chen et al., 2024), is a multilingual embedding model optimized for dense retrieval and multi-vector tasks. Built on XLM-RoBERTa (Conneau et al., 2019), it benefits from self-knowledge distillation and optimized batching. Its ability to process up to 8,192 tokens makes it ideal for compact abstracts and detailed occupation descriptions. While other models (e.g., BERT (Devlin et al., 2019), XLM- RoBERTa) were tested, BGE-M3 showed superior performance and was chosen for final analysis.

3.3.1 Patent-to-Occupation Mapping via Embedding Similarity Threshold

The final step assigns patents to occupations based on semantic similarity between patent abstracts and ISCO-08 descriptions. Using the BGE-M3 embedding model, both texts are converted into vectors, and pairwise **dot product scores** (0 to 1) semantic similarity scores are calculated. To determine whether a given patent can be reliably associated with an occupation, a similarity threshold must be defined: a minimum value above which the semantic match is considered valid. A sample dataset was analyzed to set the threshold and the results were manually reviewed. A similarity score of 0.55 was observed to provide a reasonable trade-off: below this value, associations were often too vague or generic, while above this cut-off, matches tended to be more meaningful and specific. Table 3 shows the distribution of patents by similarity score and the average number of occupational classes per patent.

Table 3. (Partial) distribution of patents with the resulting pairwise patent-occupation similarity.

Similarity	# of patents (= cutoff)	# of patents (>= cutoff)	avg # of classes per patent
0.49	330,369	1,137,945	11
0.50	248,813	807,576	8
0.51	175,254	558,763	5
0.52	131,014	383,509	4
0.53	86,151	252,495	2
0.54	61,054	166,344	2
0.55	**38,999**	**105,290**	**1**
0.56	26,104	66,291	1
0.56 +	40,187	40,187	0

Patent–occupation associations with a similarity score below 0.55 were discarded, ensuring only clear work-related links remained. The valid associations formed the final dataset for analysis. Each entry includes three key variables: **patent publication year**, **occupation code and title** (from ISCO-08 Minor group classification), and the **number of associated patents** per year. This structure creates a time series dataset, tracking AI-related innovations across occupations and identifying trends in exposure.

4. Results

The results are organized into two parts. First, we **rank occupations** most exposed to AI based on the total number of patents associated with each ISCO category. This offers a static, cumulative view of occupational relevance across the dataset. Second, we conduct a **temporal analysis**, examining how patent activity related to AI evolves for the top-ranked occupational groups. This dynamic perspective helps to identify trends and shifts in the impact of AI across the labor market.

4.1. Ranking of Occupations Most Exposed to AI

Table 4 ranks **Software and Applications Developers and Analysts** first, with 8,675 patents, highlighting software engineering's key role in AI. Next are **Database and Network Professionals** (6,317) and **Keyboard Operators** (6,034), differing by ~5% but trailing the top category by 37–40%. **Client Information Workers** (5,045) and **ICT Service Managers** (4,829) follow closely (~4% gap), though both fall ~20% below the third rank. Other notable AI-exposed groups include **Process Control Technicians**, **ICT Operations and User Support Technicians**, and even **Creative and Performing Artists**, reflecting generative AI's impact on the arts.

Table 4. Top 10 ranking of the most ISCO professions exposed to AI topics according to the number of patents associated.

ISCO Title	# of Patents	% diff from upper pos.
SW and Applications Devs and Analysts	8,675	-
Database and Network Professionals	6,317	37%
Keyboard Operators	6,034	5%
Client Information Workers	5,045	20%
ICT Service Managers	4,829	4%
Process Control Technicians	3,580	35%
ICT Oper. and User Support Technicians	3,351	7%
Creative and Performing Artists	3,297	2%
Telecomm. and Broadcasting Technicians	3,210	3%
Engineering Professionals	3,109	3%
Total	47,447	-

This ranking highlights the concentration of AI exposure in ICT-related professions and hints at AI's growing presence in non-traditional domains such as the arts and broadcasting.

4.2. Trends in Occupational Exposure to AI

Figure 1 shows the yearly distribution of AI-related patents across the 10 most exposed ISCO-08 occupations, which account for 47,447 patents (45% of the total 105,290).

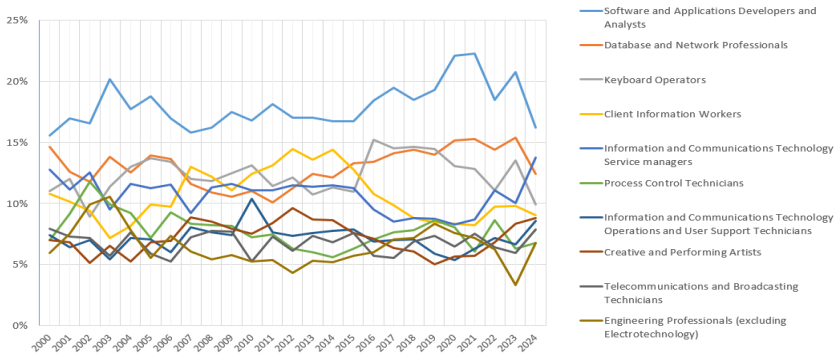

Figure 1. Time series of the top 10 ISCO professions by number of patents on AI. Percentage data on an annual basis.

The trends reveal several key patterns: **Software and Applications Developers and Analysts** lead AI-related patent activity, peaking at over 20% from 2020–2022 before a slight decline. **Database and Network Professionals** show steady growth, ranking second in AI exposure since 2011. **Keyboard Operators** see growth (2002–2005, 2015–2016) before a sharp decline, reflecting AI-driven automation replacing manual data entry. **Client Information Workers** double their AI-related patents from 2003 to 2014, then gradually decline, suggesting AI adoption peaked in customer service. **ICT Service Managers** experience fluctuations, with a sharp AI-driven surge around 2023. **Process Control Technicians** steadily increase in AI exposure, highlighting industrial automation integration. **ICT Operations and User Support Technicians** fluctuate, peaking around 2010 due to AI's role in IT support. **Creative and Performing Artists** see rising AI-related patents, nearing 10% in 2012 and increasing again recently, reflecting AI's role in generative content creation. **Telecommunications and Broadcasting Technicians** remain stable with modest AI involvement, likely due to automation in media transmission. **Engineering Professionals** peak in AI patents around 2002–2003 (>10%) before declining in 2023–2024, indicating early AI research later specialized in specific fields. These trends illustrate AI's dominance in software-related professions and its gradual penetration into less conventional domains such as creative industries and process control. The rise and decline of AI exposure in some occupations also highlights the dynamic nature of technological adoption, where initial enthusiasm may be followed by stabilization or even displacement due to automation.

5. Conclusion and Future Works

This study assessed occupational exposure to AI using LLMs, patent data, and ISCO-08 classifications. By analysing patents, we identified professions most affected by AI-related patents and tracked their evolution over time.

Results show that ICT-related roles have the highest AI exposure due to their direct involvement in AI development and deployment. Managerial and technical roles also show significant exposure, reflecting AI's growing role in automation and decision-making.

Future research will integrate labor market indicators like employment trends and wages to better assess AI's workforce impact. Additionally, refining similarity measurements and exploring alternative embedding models will improve assessment accuracy.

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